

ddRAD-seq, End to End: What It Measures, How to Execute It, and When to Use It

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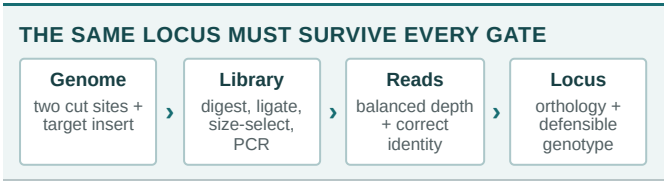
Summary

Double-digest restriction-site-associated DNA sequencing (ddRAD-seq) uses two restriction enzymes, adapters and a fragment-size window to reproducibly sequence a small genomic subset across many individuals. Its economy comes from exchanging genome continuity for sample number and locus depth. That exchange is powerful for population structure, kinship, linkage and broad differentiation, but only when field sampling, digestion, size selection, PCR, sequencing and locus assembly are treated as one coupled sampling process. Restriction-site polymorphism, degraded DNA, paralogs, batch imbalance and filter choice can otherwise convert technical absence into apparent biology. A pilot, controls, replicate concordance and inference-specific sensitivity analyses are therefore part of the experiment—not optional cleanup.

What ddRAD-seq actually measures

ddRAD-seq is reduced-representation genotyping, not low-coverage whole-genome sequencing. Genomic DNA is cut simultaneously with two sequence-specific endonucleases, commonly a rare and a frequent cutter. Asymmetric adapters recognize the two overhangs; PCR and size selection enrich fragments with the intended end combination and insert length. Sequencing reads then repeatedly sample sequence adjacent to those cut sites. Enzyme pair and size window tune the number of loci from hundreds to hundreds of thousands, while multiplexing moves sequencing effort from genome breadth to individuals [1][2].

The resulting marker set is conditional. A locus exists in the data only if both recognition sites exist in that chromosome, digestion and ligation succeed, the fragment enters the selected size distribution, it amplifies, receives reads, assembles or maps, and passes filters. Thus “missing genotype” can mean insufficient depth, whereas “missing locus” can reflect a cut-site mutation, structural change, DNA degradation or technical failure. These mechanisms must not be collapsed into one missing-data percentage.



Engineer the library on paper first

Define whether the estimand is assignment, kinship, F_{ST} , diversity, an SFS, linkage, association or an outlier signal. Allocate biological replication before maximizing SNP count; balance groups across plates, pools and lanes, and include blanks, a repeated reference and replicates that estimate discordance.

With a focal or close reference, perform an in-silico digest: count fragments with the intended ends and insert window, then inspect genomic distribution, repeats, GC and mappability. Motif length alone does not determine cutting; genome composition, methylation sensitivity and repeats matter. Without a reference, use genome-size/GC priors cautiously and run an empirical enzyme-by-window pilot. Short fragments can leak through nominal size selection, so enzyme choice may matter more than a narrower instrument setting [3].

Expected reads divided by target fragments is only nominal depth; unequal pooling, duplicates and locus-specific amplification consume it. Choose a locus target that leaves enough usable depth and samples after attrition. Before the full cohort, test fragment profile, index balance, per-sample yield, allelic balance and shared-locus saturation in a shallow pilot.

1 · Collect and qualify biological material

- Sampling frame.** Sample the units required by the question; avoid relatives unless intentional. Record location, date, tissue, life stage, sex, phenotype, preservative, storage and freeze–thaw history. Secure ethics and permits.
- Preservation.** Rapidly apply one validated cold-chain, ethanol/salt or tissue-specific regime. Use clean instruments, field blanks where relevant and unambiguous sample-chain identifiers.
- Extraction and QC.** Extract consistently with blanks. Quantify double-stranded DNA fluorometrically; assess integrity by gel/fragment analysis and matrix-appropriate purity or inhibition. Normalize amplifiable mass, not absorbance alone. Degradation preferentially feeds short-fragment pools and can erase loci before digestion.
- Acceptance.** Predefine mass, concentration, integrity and contamination thresholds. Re-extract failures where possible; otherwise distribute and flag marginal extracts for sensitivity analysis.

2 · Digest, ligate and preserve identity

1. **Plate plan and controls.** Freeze a machine-readable sample sheet before pipetting. Place blanks and repeated reference DNA across plates. Use the same DNA mass and reaction volume, validated enzyme excess, compatible buffer, incubation and inactivation/cleanup. A test digest can reveal gross inhibition; it cannot prove equal cutting at every locus.
2. **Double digest.** Both enzymes must cut efficiently under the same conditions and produce adapter-compatible ends. Star activity, partial digestion, methylation sensitivity and DNA contaminants change the sampled genome. Keep enzyme lots and timing consistent; record reagent lots, operator, thermocycler and plate.
3. **Adapter ligation.** Ligate end-specific adapters at a molar ratio validated against input fragments. Inline barcodes and external indexes must be platform-compatible, sufficiently separated in sequence and uniquely assigned. Confirm whether index reads and custom sequencing primers are required. Index design is part of sample identity; retain the exact adapter/barcode sequences.
4. **Pooling.** Pool only after each sample has an identifying barcode. Equal DNA input does not guarantee equal library yield, so choose and document pre- or post-PCR normalization. Keep biological groups balanced across pools and lanes; a population that equals a library batch is statistically inseparable from batch.

3 · Select fragments and amplify sparingly

Specify whether the reported range is insert length or total library length including adapters. Automated electrophoretic selection is often more precise than manual gel excision, but empirical distributions can remain broad and admit short fragments. Inserts shorter than the combined paired-read span cause read-through and overlap; overlap may aid consensus after merging, but severe short-fragment carryover wastes cycles on adapters and a small subset of loci [3][4].

PCR enriches correctly ligated molecules and completes platform constructs. Use a high-fidelity polymerase, the lowest validated cycle number and, when practical, parallel reactions subsequently combined. Excess cycles increase jackpots, heteroduplexes and duplicates. Fragment length and GC content can drive unequal amplification, so uniform-looking library concentration does not imply uniform locus coverage [4]. Purify without shifting the intended distribution.

4 · Library QC and sequencing

Quantify amplifiable library and verify its fragment distribution. Reject adapter dimers and unexplained shoulders; re-pool by molarity if possible. Sequence controls. Confirm read length, index orientation, custom primers and restriction-remnant handling. Early-cycle base diversity may be low; use the facility's validated loading strategy.

5 · Start bioinformatics with identity and yield

1. **Preserve provenance.** Retain raw BCL/FASTQ, checksums, sample sheet, adapter/barcode file, lane and library IDs, protocol and software environments. Version the workflow and parameters; never overwrite raw demultiplexed reads.
2. **Demultiplex.** Assign reads using the planned barcode mismatch rule, orient reads correctly, and require the expected restriction remnant where appropriate. Report assigned, rescued, ambiguous and unassigned reads per sample. Inspect blanks, index imbalance and whether low-yield samples align with plate, extraction or biological groups.
3. **Read QC.** Inspect per-base quality, length, adapter/read-through, GC, duplication and overrepresented motifs by sample and batch. Trim verified adapter sequence and unusable tails; do not apply aggressive quality trimming reflexively because read length affects overlap, mapping and de-novo clustering. Merge overlapping pairs only with a documented error-aware rule.

6 · Establish homologous loci

Reference-guided route. Map to a close, sufficiently contiguous reference; retain alignment scores and alternate mappings, and mask or filter repeats and low-mappability regions. The reference supplies genomic position, helps separate paralogs and enables LD/window analyses. It does not eliminate reference bias. In an empirical comparison of 312 analysis combinations, close-reference workflows often agreed better with external expectations, while increasing reference divergence produced signals consistent with miscalling [5].

De-novo route. When no defensible reference exists, cluster reads into within-individual stacks, merge alleles, then construct a cross-individual catalogue using tools such as Stacks, ipyrad or dDocent. Similarity thresholds govern the split-merge trade-off: too strict separates alleles; too permissive merges paralogs. Optimize the main mismatch/coverage parameters across a grid and inspect locus number, polymorphism, heterozygosity, SNPs per locus and replicate/Mendelian error. Optimal settings differed among three empirical datasets, which rules out a universal default [6]. Paired-end locus assembly can improve contiguity and orthology assessment [7][8].

7 · Call genotypes before filtering conclusions

Call variants jointly within the intended analysis cohort, or use genotype likelihoods when depth is too low or uneven for reliable hard calls. Evaluate base and mapping quality, strand/orientation, minimum and excessive depth, allelic balance, local haplotypes and indels. Separate invariant callable sites from absent loci when an SFS or diversity denominator is needed. A VCF containing only variable sites cannot by itself provide an unbiased genomic denominator.

8 · Filter by mechanism, then test robustness

Filter individuals for contamination, yield, depth and missingness before loci. Then examine group-wise call rate, minor-allele count, genotype/depth quality, extreme depth, allelic balance, heterozygosity, justified HWE tests and linkage. Global call-rate filters may preferentially erase loci absent from a divergent population; HWE filters may erase genuine structure, inbreeding or selection.

Collapsed paralogs often show excess depth or heterozygosity, more than two diploid haplotypes, or non-diploid allelic ratios; combine these diagnostics, especially after genome duplication [9]. LD-prune or keep one SNP per locus when approximate independence is required; retain and model linkage for linkage/QTL or haplotype analyses.

A cut-site mutation causes nonrandom allele/locus dropout. Its bias to diversity and genealogy depends on diversity, demography and selection [10][11]. Repeat core analyses across defensible depth, call-rate, clustering and paralog settings; test whether batch predicts PCs, phenotype or missingness. Validate with replicates, pedigrees, geography or independent markers. Inferential stability—not maximum SNP count—is the endpoint.

When ddRAD-seq is—and is not—the right instrument

Question	Fit	Condition or better alternative
Structure, assignment, kinship	Strong	Many individuals; shared loci and replicate-aware QC.
Linkage/QTL, introgression	Strong–conditional	Match marker spacing to pedigree and recombination.
Demography, SFS, diversity	Conditional	Model callable sites/dropout; prefer WGS for absolute estimates.
Selection scan or GWAS	Conditional	Density, LD and architecture govern power; sparse loci miss regions [12].
Deep/divergent phylogeny	Weak	Cut-site loss; fixed-locus capture is more portable.
Rare/SV/CNV/causal variant	Poor	Use WGS, long reads or targeted capture.

Best fit: numerous samples, expensive deep WGS, and a question answered by distributed markers. Re-price against low-pass WGS when a good reference and imputation panel exist; include library labor, failed pilots, discarded reads and limited cross-study portability.

Perspective

ddRAD-seq succeeds when the restriction library is designed as a population-genetic measurement system. The defensible endpoint is not a large VCF; it is a result that survives controls, replication and plausible analytical choices.

How many samples? First identify the denominator

No defensible ddRAD design starts with “sample a fixed percentage of the stock.” Four quantities are different: **census size** (N , all prawns present), **effective population size** (N_e , the genetic contribution of breeders), **independent units** (populations, rivers, ponds, families or time points), and **individuals genotyped within each unit**. The biological question decides which is the denominator.

For a large population, genotyping 30 rather than 20 individuals can improve allele-frequency precision, but increasing a pond from 10,000 to 100,000 prawns does not make the required sample ten times larger. Once the sampled fraction is small, uncertainty depends mainly on allele frequency, relatedness, heterogeneity, usable loci, depth and missingness. A finite-population correction becomes relevant when a simple random sample is a non-trivial fraction of a small closed stock. If an infinite-population calculation gives n_0 , the survey correction is $n = n_0/[1 + (n_0-1)/N]$. This reduces—not increases—the required n as the sampled fraction approaches the whole stock; it does not correct family structure or substitute for genetic power simulation.

Ponds, prawns and pseudoreplication

If feed, density, aeration or another treatment is assigned to an entire pond, the **pond is the experimental unit**. Fifty prawns from one treated pond provide precise information about that pond but still constitute one treatment replicate. More prawns within a pond cannot replace independent ponds; treatments require randomized replicate ponds and analysis that models prawns nested within ponds [16][17].

For a genetic stock survey with no pond-level treatment, ponds may instead be sampling strata. Sample several ponds per hatchery or stock, then several unrelated prawns per pond. Report both numbers. Avoid sampling one full-sib family produced by a few brooders: a large headcount can represent a very small N_e . For rivers, repeat this logic across reaches or collection events; for temporal monitoring, sample comparable cohorts at biologically justified intervals.

If the estimand is a single farm-wide allele frequency, allocation across ponds may be approximately proportional to prawn abundance, with inclusion probabilities retained for weighting. If the aim is to compare ponds or stocks, a more balanced allocation generally protects per-group precision. Pond area is useful only insofar as it predicts abundance or environmental heterogeneity; it is not itself the genetic denominator.

Planning ranges by inferential objective

Objective	Planning target	Critical unit
Enzyme/window pilot	12–16 total	All intended stocks and DNA-quality classes represented
Broad structure or F_{ST}	Prefer 30–40 retained per population; 20 is a fragile floor	Unrelated individuals within each biological population
Weak structure, admixture, diversity	40–60 per population	Balanced sites, cohorts and sexes where relevant
Temporal/LD-based N_e	30–60+ per population per time point	Estimator-specific simulation; rare-allele and missingness sensitivity
Parentage	All plausible parents; often hundreds of offspring	Complete candidate-parent set and families
QTL/GWAS	QTL often 100–300+ informative offspring; GWAS hundreds–thousands	Effect-size/LD power, phenotype balance and family structure

These are planning ranges, not universal minima. A two-population plant resampling study obtained stable selected estimates above eight individuals with $\geq 1,000$ SNPs, but its strong separation, taxon and endpoints do not license $n = 8$ for prawns [13]. Effective-size analyses can remain sensitive at much larger n and to SNP filters [15].

A worked udang galah sampling design

Suppose the objective is to estimate differentiation and admixture among four Malaysian *Macrobrachium rosenbergii* river or hatchery stocks. A defensible starting target is **30–40 unrelated, retained individuals per stock** (120–160 analysed). If 15–25% may fail collection, extraction, library or read thresholds, collect approximately **36–50 per stock**. Sample across relevant reaches or several ponds rather than taking siblings from one spawning event. Use the same tissue and comparable life stage, season and size class across stocks; record sex and family information where known.

Design the hierarchy, not only the headcount

For each production stock, record the total stock estimate, number and size of ponds, stocking source, broodstock number, spawning batches and transfers among ponds. These variables describe representativeness and dependence; pond surface area alone does not determine genetic n . If the question compares four pond-management treatments, prioritize several independent ponds per treatment and genotype a prespecified subsample within every pond. If the question compares four genetic stocks housed across ponds, sample every stock across multiple ponds and fit stock and pond at their correct hierarchical levels.

Let pilot data determine the final number

1. State the smallest biologically important F_{ST} , assignment accuracy, ancestry proportion, kinship difference or phenotype effect.
2. Use the pilot to estimate retained individuals, independent loci, depth distribution, nonrandom missingness, relatedness and genotyping error.
3. Simulate or repeatedly downsample candidate designs by population and pond; preserve the observed imbalance and family structure.
4. Choose n from bias, confidence-interval width, classification error and robustness across filters—not nominal significance alone.
5. Inflate collection targets for expected attrition and reserve 5–10% cross-library replicates, plus a common reference sample in every batch.

What the existing prawn study teaches

Alam et al. began with 83 wild-origin prawns from four Bangladeshi rivers (20, 20, 22 and 21). Ten individuals fell below the read threshold; ddRAD analysis retained 73 individuals, 141 stacks and 106 SNPs at mean depth 18.6. The data supported a broad Naf River separation, but they do not establish approximately 20 animals as a universal requirement. Naf samples were postlarvae, also showing why site should not be confounded with life stage [14].

Sampling principle

Count independence before headcount. A well-powered ddRAD study samples the biological hierarchy needed for its inference, then sequences enough individuals at each level to survive attrition and analytical sensitivity tests.

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Scope

Targeted searches of PubMed/PMC, DOI/publisher records and citation lists through 13 September 2026 prioritized the original method, controlled laboratory evaluations, sampling studies, pipeline comparisons and boundary evidence. This perspective is not a systematic review. Exact searches and a claim ledger accompany the paper.